

EduSaaS Full CREATE TABLE DDL Scripts

Source: baseline Edu_saas_DB_Scripts.sql plus approved backend-owned AI support tables.

Baseline CREATE TABLE count extracted from SQL file: 41

Additional AI support table count added from requirement text: 5

Total table DDLs in this document: 46

Note: The baseline SQL defines foreign keys and indexes after the CREATE TABLE blocks. Those baseline ADD FOREIGN KEY and CREATE INDEX statements are included in appendices. The five AI-support CREATE TABLE statements include their own key constraints and recommended indexes/triggers are included after them.

1. Schema and Enum DDL

```
CREATE SCHEMA IF NOT EXISTS "education";
CREATE TYPE "education"."community_post_status" AS ENUM ('Draft', 'Published', 'Archived', 'Flagged');
CREATE TYPE "education"."community_post_type" AS ENUM ('Discussion', 'Question', 'Course', 'Announcement', 'Job', 'Internship', 'Project', 'Achievement', 'Event');
CREATE TYPE "education"."community_reaction_type" AS ENUM ('Like', 'Celebrate', 'Support', 'Insightful');
CREATE TYPE "education"."community_visibility" AS ENUM ('Public', 'Students', 'Educators', 'Employers', 'Admins');
```

2. Baseline CREATE TABLE DDL from Attached SQL File

1. education.users

```
CREATE TABLE "education"."users" (
    "id" UUID NOT NULL DEFAULT gen_random_uuid(),
    "name" VARCHAR(255) NOT NULL,
    "email" VARCHAR(255) NOT NULL,
    "role_id" UUID NOT NULL,
    "password_hash" TEXT NOT NULL,
    "status" VARCHAR(50) NOT NULL DEFAULT 'active',
    "last_login" TIMESTAMP(6),
    "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "domain_role_id" UUID,

    CONSTRAINT "users_pkey" PRIMARY KEY ("id")
);
```

2. education.roles

```
CREATE TABLE "education"."roles" (
    "id" UUID NOT NULL DEFAULT gen_random_uuid(),
    "name" VARCHAR(100) NOT NULL,
    "description" TEXT,
    "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
```

CONSTRAINT "roles_pkey" PRIMARY KEY ("id")

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```
);
```

3. education.permissions

```
CREATE TABLE "education"."permissions" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "name" VARCHAR(150) NOT NULL,  
  "description" TEXT,  
  "category" VARCHAR(100),  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "permissions_pkey" PRIMARY KEY ("id")  
);
```

4. education.role_permissions

```
CREATE TABLE "education"."role_permissions" (  
  "role_id" UUID NOT NULL,  
  "permission_id" UUID NOT NULL,  
  "assigned_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "role_permissions_pkey" PRIMARY KEY ("role_id","permission_id")  
);
```

5. education.profiles

```
CREATE TABLE "education"."profiles" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "career_goal" TEXT,  
  "institution" TEXT,  
  "company" TEXT,  
  "preferences" JSONB,  
  "initial_assessment_completed" BOOLEAN NOT NULL DEFAULT false,  
  
  CONSTRAINT "profiles_pkey" PRIMARY KEY ("id")  
);
```

6. education.assessments

```
CREATE TABLE "education"."assessments" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "type" VARCHAR(100) NOT NULL,  
  "score" DOUBLE PRECISION NOT NULL,  
  "answers" JSONB,  
  "date_taken" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "career_goal" TEXT,  
  "is_initial" BOOLEAN NOT NULL DEFAULT false,  
  "total_marks" INTEGER,  
  "obtained_marks" INTEGER,  
  "completed" BOOLEAN NOT NULL DEFAULT false,  
  "started_at" TIMESTAMP(6),  
  "completed_at" TIMESTAMP(6),  
  
  CONSTRAINT "assessments_pkey" PRIMARY KEY ("id")  
);
```

7. education.gap_reports

```
CREATE TABLE "education"."gap_reports" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "readiness_score" DOUBLE PRECISION NOT NULL,  
  "missing_skills" TEXT[],  
  "recommendations" JSONB,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "gap_reports_pkey" PRIMARY KEY ("id")  
);
```

8. education.courses

```
CREATE TABLE "education"."courses" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "title" VARCHAR(255) NOT NULL,  
  "description" TEXT,  
  "provider" VARCHAR(255),  
  "category" VARCHAR(255),  
  "difficulty" VARCHAR(50),  
  "status" VARCHAR(50) NOT NULL DEFAULT 'active',  
  "educator_id" UUID,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "courses_pkey" PRIMARY KEY ("id")  
);
```

9. education.enrollments

```
CREATE TABLE "education"."enrollments" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "course_id" UUID NOT NULL,  
  "status" VARCHAR(50) NOT NULL DEFAULT 'active',  
  "completion_percentage" DOUBLE PRECISION NOT NULL DEFAULT 0,  
  "enrolled_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "enrollments_pkey" PRIMARY KEY ("id")  
);
```

10. education.jobs

```
CREATE TABLE "education"."jobs" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "employer_id" UUID NOT NULL,  
  "title" VARCHAR(255) NOT NULL,  
  "description" TEXT,  
  "requirements" TEXT,  
  "required_skills" TEXT[],  
  "status" VARCHAR(50) NOT NULL DEFAULT 'open',  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "jobs_pkey" PRIMARY KEY ("id")  
);
```

11. education.applications

```
CREATE TABLE "education"."applications" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "job_id" UUID NOT NULL,  
  "student_id" UUID NOT NULL,  
  "status" VARCHAR(50) NOT NULL DEFAULT 'submitted',  
  "skill_match" DOUBLE PRECISION,  
  "applied_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "applications_pkey" PRIMARY KEY ("id")  
);
```

12. education.notifications

```
CREATE TABLE "education"."notifications" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "type" VARCHAR(100) NOT NULL,  
  "message" TEXT NOT NULL,  
  "read_status" BOOLEAN NOT NULL DEFAULT false,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "notifications_pkey" PRIMARY KEY ("id")  
);
```

13. education.subscriptions

```
CREATE TABLE "education"."subscriptions" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "plan_type" VARCHAR(100) NOT NULL,  
  "start_date" TIMESTAMP(6) NOT NULL,  
  "end_date" TIMESTAMP(6) NOT NULL,  
  
  CONSTRAINT "subscriptions_pkey" PRIMARY KEY ("id")  
);
```

14. education.reports

```
CREATE TABLE "education"."reports" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "title" VARCHAR(255) NOT NULL,  
  "type" VARCHAR(100) NOT NULL,  
  "generated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "exported_at" TIMESTAMP(6),  
  "format" VARCHAR(50),  
  "payload" JSONB,  
  
  CONSTRAINT "reports_pkey" PRIMARY KEY ("id")  
);
```

15. education.settings

```
CREATE TABLE "education"."settings" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "scope" VARCHAR(100) NOT NULL DEFAULT 'system',  
  "key" VARCHAR(255) NOT NULL,  
  "value" JSONB NOT NULL,
```

```
CONSTRAINT "settings_pkey" PRIMARY KEY ("id")
);
```

16. education.lessons

```
CREATE TABLE "education"."lessons" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "course_id" UUID NOT NULL,
  "title" VARCHAR(255) NOT NULL,
  "video_url" TEXT,
  "duration" INTEGER,
  "order_index" INTEGER NOT NULL DEFAULT 0,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "lessons_pkey" PRIMARY KEY ("id")
);
```

17. education.quizzes

```
CREATE TABLE "education"."quizzes" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "lesson_id" UUID NOT NULL,
  "questions" JSONB NOT NULL,
  "passing_score" INTEGER NOT NULL DEFAULT 60,

  CONSTRAINT "quizzes_pkey" PRIMARY KEY ("id")
);
```

18. education.assignments

```
CREATE TABLE "education"."assignments" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "lesson_id" UUID NOT NULL,
  "instructions" TEXT,
  "submission_link" TEXT,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "assignments_pkey" PRIMARY KEY ("id")
);
```

19. education.progress

```
CREATE TABLE "education"."progress" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "lesson_id" UUID NOT NULL,
  "watched_duration" INTEGER NOT NULL DEFAULT 0,
  "quiz_score" INTEGER,
  "assignment_status" VARCHAR(50) NOT NULL DEFAULT 'pending',
  "completion_flag" BOOLEAN NOT NULL DEFAULT false,
  "updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "completed_at" TIMESTAMP(6),

  CONSTRAINT "progress_pkey" PRIMARY KEY ("id")
);
```

20. education.certificates

```
CREATE TABLE "education"."certificates" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "course_id" UUID NOT NULL,  
  "certificate_code" VARCHAR(255) NOT NULL,  
  "issued_date" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "certificates_pkey" PRIMARY KEY ("id")  
);
```

21. education.achievements

```
CREATE TABLE "education"."achievements" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "badge_name" VARCHAR(255) NOT NULL,  
  "milestone" TEXT,  
  "certificate_id" UUID,  
  "earned_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "achievements_pkey" PRIMARY KEY ("id")  
);
```

22. education.tasks

```
CREATE TABLE "education"."tasks" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "course_id" UUID,  
  "title" VARCHAR(255) NOT NULL,  
  "due_date" TIMESTAMP(6),  
  "status" VARCHAR(50) NOT NULL DEFAULT 'pending',  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  "completed_at" TIMESTAMP(3),  
  
  CONSTRAINT "tasks_pkey" PRIMARY KEY ("id")  
);
```

23. education.recommendations

```
CREATE TABLE "education"."recommendations" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "user_id" UUID NOT NULL,  
  "course_id" UUID NOT NULL,  
  "reason" TEXT,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "recommendations_pkey" PRIMARY KEY ("id")  
);
```

24. education.announcements

```
CREATE TABLE "education"."announcements" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "educator_id" UUID NOT NULL,  
  "title" VARCHAR(255) NOT NULL,  
  "message" TEXT NOT NULL,
```

```

"audience" VARCHAR(100) NOT NULL DEFAULT 'all',
"scheduled_at" TIMESTAMP(6),
"attachment" TEXT,
"created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "announcements_pkey" PRIMARY KEY ("id")
);

```

25. education.auth_otps

```

CREATE TABLE "education"."auth_otps" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "otp_hash" TEXT NOT NULL,
  "expires_at" TIMESTAMP(6) NOT NULL,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "verified_at" TIMESTAMP(6),

  CONSTRAINT "auth_otps_pkey" PRIMARY KEY ("id")
);

```

26. education.refresh_tokens

```

CREATE TABLE "education"."refresh_tokens" (
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "user_id" UUID NOT NULL,
  "token_hash" TEXT NOT NULL,
  "expires_at" TIMESTAMP(6) NOT NULL,
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "refresh_tokens_pkey" PRIMARY KEY ("id")
);

```

27. education.domain_roles

```

CREATE TABLE "education"."domain_roles" (
  "domain_role_id" UUID NOT NULL DEFAULT gen_random_uuid(),
  "domain_name" VARCHAR(100) NOT NULL,
  "category" VARCHAR(50),
  "created_at" TIMESTAMP(6) DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "domain_roles_pkey" PRIMARY KEY ("domain_role_id")
);

```

28. education.skills

```

CREATE TABLE "education"."skills" (
  "skill_id" SERIAL NOT NULL,
  "skill_name" VARCHAR(100) NOT NULL,
  "category" VARCHAR(50),
  "description" TEXT,
  "created_at" TIMESTAMP(6) DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "skills_pkey" PRIMARY KEY ("skill_id")
);

```

29. education.domain_required_skills

```
CREATE TABLE "education"."domain_required_skills" (  
    "domain_required_skill_id" SERIAL NOT NULL,  
    "domain_role_id" UUID NOT NULL,  
    "skill_id" INTEGER NOT NULL,  
    "required_level" SMALLINT NOT NULL,  
  
    CONSTRAINT "domain_required_skills_pkey" PRIMARY KEY ("domain_required_skill_id")  
);
```

30. education.difficulty_levels

```
CREATE TABLE "education"."difficulty_levels" (  
    "difficulty_id" SERIAL NOT NULL,  
    "difficulty_name" VARCHAR(20) NOT NULL,  
    "difficulty_order" INTEGER NOT NULL,  
  
    CONSTRAINT "difficulty_levels_pkey" PRIMARY KEY ("difficulty_id")  
);
```

31. education.questions

```
CREATE TABLE "education"."questions" (  
    "question_id" SERIAL NOT NULL,  
    "skill_id" INTEGER NOT NULL,  
    "difficulty_id" INTEGER NOT NULL,  
    "question_text" TEXT NOT NULL,  
    "option_a" TEXT NOT NULL,  
    "option_b" TEXT NOT NULL,  
    "option_c" TEXT NOT NULL,  
    "option_d" TEXT NOT NULL,  
    "correct_option" CHAR(1) NOT NULL,  
    "marks" INTEGER DEFAULT 1,  
    "is_active" BOOLEAN DEFAULT true,  
    "created_at" TIMESTAMP(6) DEFAULT CURRENT_TIMESTAMP,  
  
    CONSTRAINT "questions_pkey" PRIMARY KEY ("question_id")  
);
```

32. education.quiz_sessions

```
CREATE TABLE "education"."quiz_sessions" (  
    "session_id" SERIAL NOT NULL,  
    "user_id" UUID NOT NULL,  
    "domain_role_id" UUID NOT NULL,  
    "start_time" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    "end_time" TIMESTAMP(6),  
    "status" VARCHAR(20) NOT NULL DEFAULT 'In Progress',  
    "total_questions" INTEGER NOT NULL DEFAULT 50,  
    "questions_answered" INTEGER NOT NULL DEFAULT 0,  
    "current_skill_id" INTEGER,  
  
    CONSTRAINT "quiz_sessions_pkey" PRIMARY KEY ("session_id")  
);
```

33. education.student_answers

```
CREATE TABLE "education"."student_answers" (  

```



```

"answer_id" SERIAL NOT NULL,
"session_id" INTEGER NOT NULL,
"skill_id" INTEGER NOT NULL,
"question_id" INTEGER NOT NULL,
"difficulty_id" INTEGER NOT NULL,
"selected_option" CHAR(1) NOT NULL,
"correct_option" CHAR(1) NOT NULL,
"is_correct" BOOLEAN NOT NULL,
"marks_awarded" INTEGER NOT NULL DEFAULT 0,
"answered_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "student_answers_pkey" PRIMARY KEY ("answer_id")
);

```

34. education.student_skill_results

```

CREATE TABLE "education"."student_skill_results" (
    "result_id" SERIAL NOT NULL,
    "session_id" INTEGER NOT NULL,
    "skill_id" INTEGER NOT NULL,
    "obtained_score" INTEGER NOT NULL,
    "maximum_score" INTEGER NOT NULL,
    "percentage" DECIMAL(5,2) NOT NULL,
    "skill_level" SMALLINT NOT NULL,
    "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "student_skill_results_pkey" PRIMARY KEY ("result_id")
);

```

35. education.quiz_state

```

CREATE TABLE "education"."quiz_state" (
    "session_id" INTEGER NOT NULL,
    "skill_id" INTEGER NOT NULL,
    "current_difficulty" INTEGER DEFAULT 1,
    "correct_streak" INTEGER DEFAULT 0,
    "wrong_streak" INTEGER DEFAULT 0,
    "questions_answered" INTEGER DEFAULT 0,
    "obtained_score" INTEGER DEFAULT 0,
    "maximum_score" INTEGER DEFAULT 0,
    "state" JSONB,

    CONSTRAINT "quiz_state_pkey" PRIMARY KEY ("session_id","skill_id")
);

```

36. education.community_bookmarks

```

CREATE TABLE "education"."community_bookmarks" (
    "id" UUID NOT NULL DEFAULT gen_random_uuid(),
    "user_id" UUID NOT NULL,
    "post_id" UUID NOT NULL,
    "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "community_bookmarks_pkey" PRIMARY KEY ("id")
);

```

37. education.community_comments

```

CREATE TABLE "education"."community_comments" (

```

```

        "id" UUID NOT NULL DEFAULT gen_random_uuid(),
        "post_id" UUID NOT NULL,
        "author_id" UUID NOT NULL,
        "parent_id" UUID,
        "content" TEXT NOT NULL,
        "reactions_count" INTEGER NOT NULL DEFAULT 0,
        "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
        "updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
        "deleted_at" TIMESTAMP(6),

        CONSTRAINT "community_comments_pkey" PRIMARY KEY ("id")
    );

```

38. education.community_post_tags

```

CREATE TABLE "education"."community_post_tags" (
    "post_id" UUID NOT NULL,
    "tag_id" UUID NOT NULL,
    "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "pk_community_post_tags" PRIMARY KEY ("post_id","tag_id")
);

```

39. education.community_posts

```

CREATE TABLE "education"."community_posts" (
    "id" UUID NOT NULL DEFAULT gen_random_uuid(),
    "author_id" UUID NOT NULL,
    "title" VARCHAR(255) NOT NULL,
    "content" TEXT NOT NULL,
    "post_type" "education"."community_post_type" NOT NULL,
    "status" "education"."community_post_status" NOT NULL DEFAULT 'Published',
    "visibility" "education"."community_visibility" NOT NULL DEFAULT 'Public',
    "media_url" TEXT,
    "metadata" JSONB,
    "comments_count" INTEGER NOT NULL DEFAULT 0,
    "reactions_count" INTEGER NOT NULL DEFAULT 0,
    "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updated_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "deleted_at" TIMESTAMP(6),

    CONSTRAINT "community_posts_pkey" PRIMARY KEY ("id")
);

```

40. education.community_reactions

```

CREATE TABLE "education"."community_reactions" (
    "id" UUID NOT NULL DEFAULT gen_random_uuid(),
    "user_id" UUID NOT NULL,
    "post_id" UUID,
    "comment_id" UUID,
    "reaction_type" "education"."community_reaction_type" NOT NULL DEFAULT 'Like',
    "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "community_reactions_pkey" PRIMARY KEY ("id")
);

```

41. education.community_tags

```
CREATE TABLE "education"."community_tags" (  
  "id" UUID NOT NULL DEFAULT gen_random_uuid(),  
  "name" VARCHAR(50) NOT NULL,  
  "slug" VARCHAR(50) NOT NULL,  
  "usage_count" INTEGER NOT NULL DEFAULT 0,  
  "created_at" TIMESTAMP(6) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
  CONSTRAINT "community_tags_pkey" PRIMARY KEY ("id")  
);
```

3. Additional Backend-Owned AI Support CREATE TABLE DDL

42. education.activity_logs

```
CREATE TABLE IF NOT EXISTS education.activity_logs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID NOT NULL,  
  sessions_last_30_days INTEGER NOT NULL,  
  avg_session_minutes DOUBLE PRECISION,  
  videos_watched INTEGER,  
  assignments_attempted INTEGER,  
  discussion_interactions INTEGER,  
  login_count INTEGER,  
  device_count INTEGER,  
  ip_changes INTEGER,  
  last_activity TIMESTAMP WITHOUT TIME ZONE,  
  suspicious_activity_score NUMERIC,  
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_activity_logs_user  
    FOREIGN KEY (user_id) REFERENCES education.users(id)  
    ON DELETE CASCADE  
);
```

43. education.login_history

```
CREATE TABLE IF NOT EXISTS education.login_history (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID NOT NULL,  
  logins_last_30_days INTEGER,  
  days_since_last_login INTEGER,  
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_login_history_user  
    FOREIGN KEY (user_id) REFERENCES education.users(id)  
    ON DELETE CASCADE  
);
```

44. education.course_skills

```
CREATE TABLE IF NOT EXISTS education.course_skills (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  course_id UUID NOT NULL,  
  skill_id INTEGER NOT NULL,  
  skill_level INTEGER,  
  created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
```

```

updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
CONSTRAINT fk_course_skills_course
    FOREIGN KEY (course_id) REFERENCES education.courses(id)
    ON DELETE CASCADE,
CONSTRAINT fk_course_skills_skill
    FOREIGN KEY (skill_id) REFERENCES education.skills(skill_id)
    ON DELETE CASCADE,
CONSTRAINT uq_course_skills_course_skill
    UNIQUE (course_id, skill_id)
);

```

45. education.course_prerequisites

```

CREATE TABLE IF NOT EXISTS education.course_prerequisites (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    course_id UUID NOT NULL,
    prerequisite_course_id UUID NOT NULL,
    created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_course_prerequisites_course
        FOREIGN KEY (course_id) REFERENCES education.courses(id)
        ON DELETE CASCADE,
    CONSTRAINT fk_course_prerequisites_prerequisite
        FOREIGN KEY (prerequisite_course_id) REFERENCES education.courses(id)
        ON DELETE CASCADE,
    CONSTRAINT uq_course_prerequisite
        UNIQUE (course_id, prerequisite_course_id),
    CONSTRAINT chk_course_prerequisite_not_same
        CHECK (course_id <> prerequisite_course_id)
);

```

46. education.course_ratings

```

CREATE TABLE IF NOT EXISTS education.course_ratings (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL,
    course_id UUID NOT NULL,
    rating DOUBLE PRECISION NOT NULL,
    created_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_course_ratings_user
        FOREIGN KEY (user_id) REFERENCES education.users(id)
        ON DELETE CASCADE,
    CONSTRAINT fk_course_ratings_course
        FOREIGN KEY (course_id) REFERENCES education.courses(id)
        ON DELETE CASCADE,
    CONSTRAINT chk_course_rating
        CHECK (rating >= 1 AND rating <= 5),
    CONSTRAINT uq_course_rating_user_course
        UNIQUE (user_id, course_id)
);

```

4. Existing recommendations Table AI ALTER Script

```

ALTER TABLE education.recommendations
    ADD COLUMN IF NOT EXISTS predicted_rating DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS similarity_score DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS confidence_score DOUBLE PRECISION,
    ADD COLUMN IF NOT EXISTS prerequisite_completed BOOLEAN,

```

```

ADD COLUMN IF NOT EXISTS rank INTEGER,
ADD COLUMN IF NOT EXISTS updated_at TIMESTAMP WITHOUT TIME ZONE DEFAULT CURRENT_TIMESTAMP;

ALTER TABLE education.recommendations
ADD CONSTRAINT chk_recommendations_predicted_rating
CHECK (predicted_rating IS NULL OR (predicted_rating >= 1 AND predicted_rating <= 5));

ALTER TABLE education.recommendations
ADD CONSTRAINT chk_recommendations_similarity_score
CHECK (similarity_score IS NULL OR (similarity_score >= 0 AND similarity_score <= 1));

ALTER TABLE education.recommendations
ADD CONSTRAINT chk_recommendations_confidence_score
CHECK (confidence_score IS NULL OR (confidence_score >= 0 AND confidence_score <= 1));

ALTER TABLE education.recommendations
ADD CONSTRAINT chk_recommendations_rank
CHECK (rank IS NULL OR rank > 0);

```

5. Baseline CREATE INDEX Statements from Attached SQL File

Total baseline index statements extracted: 76

```

CREATE UNIQUE INDEX "users_email_key" ON "education"."users"("email");
CREATE INDEX "idx_users_role" ON "education"."users"("role_id");
CREATE INDEX "users_domain_role_id_idx" ON "education"."users"("domain_role_id");
CREATE UNIQUE INDEX "roles_name_key" ON "education"."roles"("name");
CREATE UNIQUE INDEX "permissions_name_key" ON "education"."permissions"("name");
CREATE UNIQUE INDEX "profiles_user_id_key" ON "education"."profiles"("user_id");
CREATE INDEX "idx_assessment_user" ON "education"."assessments"("user_id");
CREATE INDEX "idx_gapreport_user" ON "education"."gap_reports"("user_id");
CREATE INDEX "idx_courses_educator" ON "education"."courses"("educator_id");
CREATE INDEX "idx_enrollment_course" ON "education"."enrollments"("course_id");
CREATE INDEX "idx_enrollment_user" ON "education"."enrollments"("user_id");
CREATE UNIQUE INDEX "uq_enrollment" ON "education"."enrollments"("user_id", "course_id");
CREATE INDEX "idx_jobs_employer" ON "education"."jobs"("employer_id");
CREATE INDEX "idx_application_job" ON "education"."applications"("job_id");
CREATE INDEX "idx_application_student" ON "education"."applications"("student_id");
CREATE UNIQUE INDEX "uq_application" ON "education"."applications"("job_id", "student_id");
CREATE INDEX "idx_notification_user" ON "education"."notifications"("user_id");
CREATE INDEX "idx_subscription_user" ON "education"."subscriptions"("user_id");
CREATE UNIQUE INDEX "uq_settings" ON "education"."settings"("scope", "key");
CREATE INDEX "idx_lessons_course" ON "education"."lessons"("course_id");
CREATE INDEX "idx_quizzes_lesson" ON "education"."quizzes"("lesson_id");
CREATE INDEX "idx_assignments_lesson" ON "education"."assignments"("lesson_id");
CREATE INDEX "idx_progress_lesson" ON "education"."progress"("lesson_id");
CREATE INDEX "idx_progress_user" ON "education"."progress"("user_id");
CREATE UNIQUE INDEX "uq_progress" ON "education"."progress"("user_id", "lesson_id");
CREATE UNIQUE INDEX "certificates_certificate_code_key" ON "education"."certificates"("certificate_code");
CREATE INDEX "idx_certificates_course" ON "education"."certificates"("course_id");
CREATE INDEX "idx_certificates_user" ON "education"."certificates"("user_id");
CREATE INDEX "idx_achievements_certificate" ON "education"."achievements"("certificate_id");
CREATE INDEX "idx_achievements_user" ON "education"."achievements"("user_id");
CREATE INDEX "idx_tasks_course" ON "education"."tasks"("course_id");
CREATE INDEX "idx_tasks_user" ON "education"."tasks"("user_id");
CREATE INDEX "idx_recommendations_course" ON "education"."recommendations"("course_id");
CREATE INDEX "idx_recommendations_user" ON "education"."recommendations"("user_id");

```

```

CREATE INDEX "idx_announcements_educator" ON "education"."announcements"("educator_id");
CREATE UNIQUE INDEX "uq_auth_otps_user_id" ON "education"."auth_otps"("user_id");
CREATE INDEX "idx_refresh_tokens_expires" ON "education"."refresh_tokens"("expires_at");
CREATE INDEX "idx_refresh_tokens_user" ON "education"."refresh_tokens"("user_id");
CREATE UNIQUE INDEX "domain_roles_domain_name_key" ON "education"."domain_roles"("domain_name");
CREATE UNIQUE INDEX "skills_skill_name_key" ON "education"."skills"("skill_name");
CREATE UNIQUE INDEX "domain_required_skills_domain_role_id_skill_id_key" ON "education"."domain_required_skills"("domain_role_id", "skill_id");
CREATE UNIQUE INDEX "difficulty_levels_difficulty_name_key" ON "education"."difficulty_levels"("difficulty_name");
CREATE UNIQUE INDEX "difficulty_levels_difficulty_order_key" ON "education"."difficulty_levels"("difficulty_order");
CREATE INDEX "idx_quiz_sessions_current_skill" ON "education"."quiz_sessions"("current_skill_id");
CREATE INDEX "idx_quiz_sessions_domain_role" ON "education"."quiz_sessions"("domain_role_id");
CREATE INDEX "idx_quiz_sessions_status" ON "education"."quiz_sessions"("status");
CREATE INDEX "idx_quiz_sessions_user" ON "education"."quiz_sessions"("user_id");
CREATE INDEX "idx_student_answers_difficulty" ON "education"."student_answers"("difficulty_id");
CREATE INDEX "idx_student_answers_question" ON "education"."student_answers"("question_id");
CREATE INDEX "idx_student_answers_session" ON "education"."student_answers"("session_id");
CREATE INDEX "idx_student_answers_skill" ON "education"."student_answers"("skill_id");
CREATE INDEX "idx_student_skill_results_session" ON "education"."student_skill_results"("session_id");
CREATE INDEX "idx_student_skill_results_skill" ON "education"."student_skill_results"("skill_id");
CREATE INDEX "idx_bookmark_post" ON "education"."community_bookmarks"("post_id");
CREATE INDEX "idx_bookmark_user" ON "education"."community_bookmarks"("user_id");
CREATE UNIQUE INDEX "uq_bookmark_user_post" ON "education"."community_bookmarks"("user_id", "post_id");
CREATE INDEX "idx_community_comments_author" ON "education"."community_comments"("author_id");
CREATE INDEX "idx_community_comments_created" ON "education"."community_comments"("created_at" DESC);
CREATE INDEX "idx_community_comments_deleted" ON "education"."community_comments"("deleted_at");
CREATE INDEX "idx_community_comments_parent" ON "education"."community_comments"("parent_id");
CREATE INDEX "idx_community_comments_post" ON "education"."community_comments"("post_id");
CREATE INDEX "idx_post_tags_post" ON "education"."community_post_tags"("post_id");
CREATE INDEX "idx_post_tags_tag" ON "education"."community_post_tags"("tag_id");
CREATE INDEX "idx_community_posts_author" ON "education"."community_posts"("author_id");
CREATE INDEX "idx_community_posts_created_at" ON "education"."community_posts"("created_at" DESC);
CREATE INDEX "idx_community_posts_deleted_at" ON "education"."community_posts"("deleted_at");
CREATE INDEX "idx_community_posts_post_type" ON "education"."community_posts"("post_type");
CREATE INDEX "idx_community_posts_status" ON "education"."community_posts"("status");
CREATE INDEX "idx_community_posts_visibility" ON "education"."community_posts"("visibility");
CREATE INDEX "idx_reaction_comment" ON "education"."community_reactions"("comment_id");
CREATE INDEX "idx_reaction_post" ON "education"."community_reactions"("post_id");
CREATE INDEX "idx_reaction_user" ON "education"."community_reactions"("user_id");
CREATE UNIQUE INDEX "uq_community_tag_name" ON "education"."community_tags"("name");
CREATE UNIQUE INDEX "uq_community_tag_slug" ON "education"."community_tags"("slug");
CREATE INDEX "idx_community_tags_slug" ON "education"."community_tags"("slug");
CREATE INDEX "idx_community_tags_usage" ON "education"."community_tags"("usage_count" DESC);

```

6. Baseline FOREIGN KEY ALTER Statements from Attached SQL File

Total baseline FK alter statements extracted: 56

```

ALTER TABLE "education"."users" ADD CONSTRAINT "fk_users_role" FOREIGN KEY ("role_id") REFERENCES "education"."roles"("id") ON DELETE NO ACTION ON UPDATE NO ACTION;
ALTER TABLE "education"."users" ADD CONSTRAINT "users_domain_role_id_fkey" FOREIGN KEY ("domain_role_id") REFERENCES "education"."domain_roles"("domain_role_id") ON DELETE SET
NULL ON UPDATE CASCADE;
ALTER TABLE "education"."role_permissions" ADD CONSTRAINT "fk_rolepermission_permission" FOREIGN KEY ("permission_id") REFERENCES "education"."permissions"("id") ON DELETE
CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."role_permissions" ADD CONSTRAINT "fk_rolepermission_role" FOREIGN KEY ("role_id") REFERENCES "education"."roles"("id") ON DELETE CASCADE ON UPDATE NO
ACTION;
ALTER TABLE "education"."profiles" ADD CONSTRAINT "fk_profile_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."assessments" ADD CONSTRAINT "fk_assessment_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."gap_reports" ADD CONSTRAINT "fk_gapreport_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."courses" ADD CONSTRAINT "fk_course_educator" FOREIGN KEY ("educator_id") REFERENCES "education"."users"("id") ON DELETE SET NULL ON UPDATE NO ACTION;

```

```

ALTER TABLE "education"."enrollments" ADD CONSTRAINT "fk_enrollment_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."enrollments" ADD CONSTRAINT "fk_enrollment_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."jobs" ADD CONSTRAINT "fk_job_employer" FOREIGN KEY ("employer_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."applications" ADD CONSTRAINT "fk_application_job" FOREIGN KEY ("job_id") REFERENCES "education"."jobs"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."applications" ADD CONSTRAINT "fk_application_student" FOREIGN KEY ("student_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."notifications" ADD CONSTRAINT "fk_notification_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."subscriptions" ADD CONSTRAINT "fk_subscription_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."lessons" ADD CONSTRAINT "fk_lesson_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."quizzes" ADD CONSTRAINT "fk_quiz_lesson" FOREIGN KEY ("lesson_id") REFERENCES "education"."lessons"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."assignments" ADD CONSTRAINT "fk_assignment_lesson" FOREIGN KEY ("lesson_id") REFERENCES "education"."lessons"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."progress" ADD CONSTRAINT "fk_progress_lesson" FOREIGN KEY ("lesson_id") REFERENCES "education"."lessons"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."progress" ADD CONSTRAINT "fk_progress_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."certificates" ADD CONSTRAINT "fk_certificate_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."certificates" ADD CONSTRAINT "fk_certificate_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."achievements" ADD CONSTRAINT "fk_achievement_certificate" FOREIGN KEY ("certificate_id") REFERENCES "education"."certificates"("id") ON DELETE SET NULL ON UPDATE NO ACTION;
ALTER TABLE "education"."achievements" ADD CONSTRAINT "fk_achievement_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."tasks" ADD CONSTRAINT "fk_task_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE SET NULL ON UPDATE NO ACTION;
ALTER TABLE "education"."tasks" ADD CONSTRAINT "fk_task_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."recommendations" ADD CONSTRAINT "fk_recommendation_course" FOREIGN KEY ("course_id") REFERENCES "education"."courses"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."recommendations" ADD CONSTRAINT "fk_recommendation_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."announcements" ADD CONSTRAINT "fk_announcement_educator" FOREIGN KEY ("educator_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."auth_otps" ADD CONSTRAINT "fk_auth_otps_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."refresh_tokens" ADD CONSTRAINT "fk_refresh_tokens_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."domain_required_skills" ADD CONSTRAINT "fk_domain_role" FOREIGN KEY ("domain_role_id") REFERENCES "education"."domain_roles"("domain_role_id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."domain_required_skills" ADD CONSTRAINT "fk_skill" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."questions" ADD CONSTRAINT "questions_difficulty_id_fkey" FOREIGN KEY ("difficulty_id") REFERENCES "education"."difficulty_levels"("difficulty_id") ON DELETE NO ACTION ON UPDATE NO ACTION;
ALTER TABLE "education"."questions" ADD CONSTRAINT "questions_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE NO ACTION ON UPDATE NO ACTION;
ALTER TABLE "education"."quiz_sessions" ADD CONSTRAINT "quiz_sessions_domain_role_id_fkey" FOREIGN KEY ("domain_role_id") REFERENCES "education"."domain_roles"("domain_role_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
ALTER TABLE "education"."quiz_sessions" ADD CONSTRAINT "quiz_sessions_user_id_fkey" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_difficulty_id_fkey" FOREIGN KEY ("difficulty_id") REFERENCES "education"."difficulty_levels"("difficulty_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_question_id_fkey" FOREIGN KEY ("question_id") REFERENCES "education"."questions"("question_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_session_id_fkey" FOREIGN KEY ("session_id") REFERENCES "education"."quiz_sessions"("session_id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."student_answers" ADD CONSTRAINT "student_answers_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
ALTER TABLE "education"."student_skill_results" ADD CONSTRAINT "student_skill_results_session_id_fkey" FOREIGN KEY ("session_id") REFERENCES "education"."quiz_sessions"("session_id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."student_skill_results" ADD CONSTRAINT "student_skill_results_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE RESTRICT ON UPDATE NO ACTION;
ALTER TABLE "education"."quiz_state" ADD CONSTRAINT "quiz_state_session_id_fkey" FOREIGN KEY ("session_id") REFERENCES "education"."quiz_sessions"("session_id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."quiz_state" ADD CONSTRAINT "quiz_state_skill_id_fkey" FOREIGN KEY ("skill_id") REFERENCES "education"."skills"("skill_id") ON DELETE NO ACTION ON UPDATE NO ACTION;

```



```

ALTER TABLE "education"."community_bookmarks" ADD CONSTRAINT "fk_bookmark_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_bookmarks" ADD CONSTRAINT "fk_bookmark_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_comments" ADD CONSTRAINT "fk_comment_author" FOREIGN KEY ("author_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_comments" ADD CONSTRAINT "fk_comment_parent" FOREIGN KEY ("parent_id") REFERENCES "education"."community_comments"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_comments" ADD CONSTRAINT "fk_comment_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_post_tags" ADD CONSTRAINT "fk_post_tag_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_post_tags" ADD CONSTRAINT "fk_post_tag_tag" FOREIGN KEY ("tag_id") REFERENCES "education"."community_tags"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_posts" ADD CONSTRAINT "fk_community_post_author" FOREIGN KEY ("author_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_reactions" ADD CONSTRAINT "fk_reaction_comment" FOREIGN KEY ("comment_id") REFERENCES "education"."community_comments"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_reactions" ADD CONSTRAINT "fk_reaction_post" FOREIGN KEY ("post_id") REFERENCES "education"."community_posts"("id") ON DELETE CASCADE ON UPDATE NO ACTION;
ALTER TABLE "education"."community_reactions" ADD CONSTRAINT "fk_reaction_user" FOREIGN KEY ("user_id") REFERENCES "education"."users"("id") ON DELETE CASCADE ON UPDATE NO ACTION;

```

7. Recommended Indexes for AI Support and Recommendations

```

CREATE INDEX IF NOT EXISTS idx_activity_logs_user_id ON education.activity_logs(user_id);
CREATE INDEX IF NOT EXISTS idx_activity_logs_last_activity ON education.activity_logs(last_activity);
CREATE INDEX IF NOT EXISTS idx_activity_logs_user_last_activity ON education.activity_logs(user_id, last_activity);
CREATE INDEX IF NOT EXISTS idx_login_history_user_id ON education.login_history(user_id);
CREATE INDEX IF NOT EXISTS idx_login_history_days_since_last_login ON education.login_history(days_since_last_login);
CREATE INDEX IF NOT EXISTS idx_course_skills_course_id ON education.course_skills(course_id);
CREATE INDEX IF NOT EXISTS idx_course_skills_skill_id ON education.course_skills(skill_id);
CREATE INDEX IF NOT EXISTS idx_course_prerequisites_course_id ON education.course_prerequisites(course_id);
CREATE INDEX IF NOT EXISTS idx_course_prerequisites_prerequisite_course_id ON education.course_prerequisites(prerequisite_course_id);
CREATE INDEX IF NOT EXISTS idx_course_ratings_user_id ON education.course_ratings(user_id);
CREATE INDEX IF NOT EXISTS idx_course_ratings_course_id ON education.course_ratings(course_id);
CREATE INDEX IF NOT EXISTS idx_course_ratings_rating ON education.course_ratings(rating);
CREATE INDEX IF NOT EXISTS idx_recommendations_rank ON education.recommendations(rank);
CREATE INDEX IF NOT EXISTS idx_recommendations_confidence_score ON education.recommendations(confidence_score);
CREATE INDEX IF NOT EXISTS idx_recommendations_user_rank ON education.recommendations(user_id, rank);

```

8. Recommended Updated_at Trigger Function and Trigger DDL

```

CREATE OR REPLACE FUNCTION education.set_updated_at()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = CURRENT_TIMESTAMP;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_activity_logs_updated_at
BEFORE UPDATE ON education.activity_logs
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();

CREATE TRIGGER trg_login_history_updated_at
BEFORE UPDATE ON education.login_history

```



```
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();
```

```
CREATE TRIGGER trg_course_skills_updated_at  
BEFORE UPDATE ON education.course_skills  
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();
```

```
CREATE TRIGGER trg_course_prerequisites_updated_at  
BEFORE UPDATE ON education.course_prerequisites  
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();
```

```
CREATE TRIGGER trg_course_ratings_updated_at  
BEFORE UPDATE ON education.course_ratings  
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();
```

```
CREATE TRIGGER trg_recommendations_updated_at  
BEFORE UPDATE ON education.recommendations  
FOR EACH ROW EXECUTE FUNCTION education.set_updated_at();
```

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